

LESSON 5-4 • SESSION 2 • PREREQUISITE PRACTICE B • SKILL 1 OF 2

Area of Triangles

Builds on Lesson 5-2 Session 1 • 6.G.A.1

TODAY'S LESSON

You will add a triangle roof to a rectangle, like $\frac{1}{2} \times 26 \times 11$ for the house front.

YOU NEED TO

Find a triangle's area with $A = \frac{1}{2} \times b \times h$, like $\frac{1}{2} \times 18 \times 7$. **You will use it in Practice Form A, problems 2, 6, and 8.**

STEPS TO FOLLOW*use these same steps on every problem*

- 1 Write** $A = \frac{1}{2} \times b \times h$ with the numbers.
- 2 Halve** the even number first.
- 3 Multiply** what is left. Write square units.

WATCH ME*the steps in action*

A roof triangle has a base of 18 ft and a height of 7 ft. Find its area.

Step 1 $A = \frac{1}{2} \times 18 \times 7$.

Step 2 Half of 18 is 9.

Step 3 $9 \times 7 = 63$, so 63 ft².

Answer **63 ft²**

TRY IT WITH HELP*fill in each step*

- 1.** Base 16 ft, height 6 ft. Find the area.

Step 1 $A = \frac{1}{2} \times \underline{\hspace{2cm}} \times \underline{\hspace{2cm}}$

Step 2 Half of 16 is $\underline{\hspace{2cm}}$

Step 3 $8 \times 6 = \underline{\hspace{2cm}}$ ft²

- 2.** Base 18 in, height 5 in. Find the area.

Step 1 $A = \frac{1}{2} \times \underline{\hspace{2cm}} \times \underline{\hspace{2cm}}$

Step 2 Half of 18 is $\underline{\hspace{2cm}}$

Step 3 $9 \times 5 = \underline{\hspace{2cm}}$ in²

YOUR TURN*use the steps on your own*

- 3.** Find each area. Use $A = \frac{1}{2} \times b \times h$.

a. $\frac{1}{2} \times 12 \times 10 = \underline{\hspace{2cm}}$

b. $\frac{1}{2} \times 26 \times 9 = \underline{\hspace{2cm}}$

c. $\frac{1}{2} \times 30 \times 7 = \underline{\hspace{2cm}}$

d. $\frac{1}{2} \times 16 \times 11 = \underline{\hspace{2cm}}$

e. $\frac{1}{2} \times 22 \times 8 = \underline{\hspace{2cm}}$

f. $\frac{1}{2} \times 9 \times 6 = \underline{\hspace{2cm}}$

Area of Regular Polygons

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TODAY'S LESSON

You will split a figure into pieces, like the 18 m patio into 5 triangles.

YOU NEED TOSplit a regular polygon into equal triangles. Find one, then multiply. **You will use it in Session 1 Practice Form A, problem 6.****STEPS TO FOLLOW***use these same steps on every problem*

- 1 Count** the sides. That is the number of center triangles.
- 2 Find** one triangle: $\frac{1}{2} \times \text{side} \times \text{apothem}$.
- 3 Multiply** one triangle by the number of triangles.

WATCH ME*the steps in action*

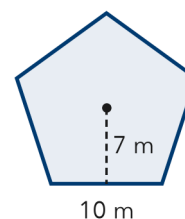
A regular pentagon has sides of 10 m and an apothem of 7 m. Find its area.

Step 1 A pentagon has 5 sides, so 5 triangles.

Step 2 $\frac{1}{2} \times 10 \times 7 = 35$.

Step 3 $5 \times 35 = 175$.

Answer **175 m²**

**TRY IT WITH HELP***fill in each step*

- 4.** A regular hexagon has sides of 8 in and an apothem of 7 in. Find its area.

Step 1 Number of triangles: _____

Step 2 $\frac{1}{2} \times 8 \times 7 =$ _____ in²

Step 3 $6 \times 28 =$ _____ in²

- 5.** A regular octagon has sides of 4 ft and an apothem of 5 ft. Find its area.

Step 1 Number of triangles: _____

Step 2 $\frac{1}{2} \times 4 \times 5 =$ _____ ft²

Step 3 $8 \times 10 =$ _____ ft²

YOUR TURN*use the steps on your own*

- 6.** Find the area of each regular polygon.

a. hexagon, side 6, apothem 5 _____

b. pentagon, side 8, apothem 6 _____

c. octagon, side 4, apothem 6 _____

d. square, side 6, apothem 3 _____

READY CHECK*a problem like the one in today's lesson*

- 7.** A 10 in by 6 in rectangle has a triangle roof: $b = 10$ in, $h = 4$ in. Total area?

A 60 in²

B 80 in²

C 100 in²

D 40 in²

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____